

CodeAlpha Data Analytics Internship

Task 2 — Exploratory Data Analysis | Task 3 — Data Visualization

Project	Ivom Retail Stores — FY2024 Sales Performance Analysis
Dataset	971-row FY2024 POS transaction export (multi-branch, multi-category retail); 956 rows after cleaning
Tool used	Microsoft Excel — PivotTables, PivotCharts, dashboard with slicer
Workbook	Sales_Analysis_Workbook.xlsx (tabs: Raw Sales Data, working sheet, pivotable, Dashboard)
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Date	September 2026

How to read this report. All figures are full-year FY2024 (January to December) totals from the cleaned 956-row dataset, with returns netted in and the workbook's Order Date slicer cleared (all months). "Revenue" means Gross Sales (NGN), the measure plotted in the workbook's charts, unless a figure is labelled Net Sales (Gross Sales less discounts). Average order value (AOV) is sales divided by the number of transactions.

TASK 2 · EXPLORATORY DATA ANALYSIS (EDA)

2.1 Meaningful questions asked before analysis

Before opening a single PivotTable, the analysis was framed around the seven business questions in the case study brief. They come from the (fictional) Regional Operations Director, who felt that "something feels off" about this year's performance but could not point to specifics.

#	Business question
1	Which branches and states are driving the most revenue, and which are underperforming?
2	Which product categories and individual products sell best, by both revenue and quantity?
3	How does monthly sales performance trend across the year? Is there seasonality?
4	How much revenue is being lost to discounts, and which customer type receives the most discount?
5	Retail vs Wholesale vs Corporate: which customer segment is most valuable to the business?
6	Which payment methods are customers actually using, and does this vary by branch?
7	Are there data quality problems in the export that could be distorting any of the figures above?

2.2 Data structure — variables and types

The raw export (sheet "Raw Sales Data") holds 971 transaction rows across 14 columns, dated 2 January to 31 December 2024. After removing 15 exact duplicate rows and standardising the text fields, 956 unique transactions remain in the "working sheet", which also carries one helper column, Discount Amount (Gross Sales less Net Sales), making 15 columns.

Column	Type	Notes
Order ID	Text (unique ID)	One row per transaction; 15 IDs appeared twice in the raw export (the duplicate rows)
Order Date	Date	Spans 2 Jan – 31 Dec 2024
Branch / State	Categorical (text)	8 branches across 5 states; the raw export held 32 different spellings/casings of the 8 branch names
Category / Product	Categorical (text)	4 categories and 24 products; the raw export held 12 spellings/casings of the 4 category names
Quantity	Numeric (integer)	6 rows had negative values (miscoded returns)
Unit Price / Gross Sales / Net Sales (NGN)	Numeric (currency)	Gross = Quantity x Unit Price (holds on every row); Net = Gross less discount
Discount (%) / Discount Amount	Numeric (percentage / currency)	Discount % takes only the values 0, 5, 10 and 15; 55 rows have no Discount % recorded

Column	Type	Notes
Customer Type	Categorical (text)	Retail, Wholesale or Corporate
Payment Method	Categorical (text)	Bank Transfer, Cash, Mobile Money or POS/Card; 21 rows missing
Salesperson	Categorical (text)	32 rows missing

2.3 Trends and patterns identified — the seven questions answered

Each of the seven questions is answered in turn below, using the cleaned "working sheet" that feeds the workbook's PivotTables. A one-line answer to every question is given first.

Q	Topic	Answer in one line
1	Branches and states	Abuja Wuse is the strongest branch (NGN 33.7M); Benue is the top state (NGN 74.7M) only because it runs 3 branches. Four branches sit below the NGN 27.6M network average; the weakest is Makurdi Main (NGN 17.5M).
2	Categories and products	Electronics earns the most revenue (NGN 79.3M); Home & Living sells the most units (1,863). Foam Mattress (6x6) is #1 by revenue and Cooking Gas Cylinder 12.5kg is #1 by quantity.
3	Monthly trend and seasonality	Volatile month to month but rising through the year: low in June (NGN 12.6M), high in December (NGN 26.8M). Q4 is the strongest quarter; a repeating seasonal pattern cannot be confirmed from one year.
4	Discounts	NGN 2,178,270 (0.99% of gross sales) is given away in discounts; Wholesale customers receive 66.8% of it.
5	Customer segments	Retail is the biggest segment by total revenue (47.0%); Wholesale is the most valuable per order (NGN 434,017 vs NGN 161,045 for Retail).
6	Payment methods	Spread almost evenly (POS/Card 26.2% to Cash 23.3% of recorded payments). Branches lean slightly differently, but the differences are not statistically significant.
7	Data quality	Yes, real problems existed (duplicates, inconsistent text, returns coded as negatives, blanks). All were handled in cleaning; none materially distorts the figures reported.

Question 1. Which branches and states are driving the most revenue, and which are underperforming?

Answer. Abuja Wuse is the strongest single branch (NGN 33.7M, 15.3% of revenue). Benue is the top state (NGN 74.7M, 33.9%), but only because it runs three of the eight branches. Four branches earn less than the NGN 27.6M network average: Makurdi Main (weakest), Lagos Lekki, Otukpo Branch and, marginally, Kano Sabon Gari.

Rank	Branch	State	Gross sales (NGN)	Share	Orders	AOV (NGN)	vs network avg
1	Abuja Wuse	FCT	33,690,100	15.3%	139	242,375	+22.2%
2	Gboko Branch	Benue	32,287,300	14.6%	133	242,762	+17.1%
3	Lagos Ikeja	Lagos	31,639,000	14.3%	129	245,264	+14.8%
4	Port Harcourt	Rivers	29,732,600	13.5%	103	288,666	+7.9%
5	Kano Sabon Gari	Kano	27,277,100	12.4%	118	231,162	-1.0%
6	Otukpo Branch	Benue	24,862,300	11.3%	121	205,474	-9.8%
7	Lagos Lekki	Lagos	23,505,000	10.7%	115	204,391	-14.7%
8	Makurdi Main	Benue	17,536,600	8.0%	98	178,945	-36.4%

State	Branches	Gross sales (NGN)	Share	Gross sales per branch (NGN)
Benue	3	74,686,200	33.9%	24,895,400
Lagos	2	55,144,000	25.0%	27,572,000
FCT	1	33,690,100	15.3%	33,690,100
Rivers	1	29,732,600	13.5%	29,732,600
Kano	1	27,277,100	12.4%	27,277,100

- **Benue leads by state total, not by branch strength.** Its three outlets average NGN 24.9M each, the lowest of the five states (FCT NGN 33.7M, Rivers NGN 29.7M, Lagos NGN 27.6M, Kano NGN 27.3M). Within Benue the spread is wide: Gboko Branch is #2 nationally (NGN 32.3M), while Makurdi Main is last.

- **Makurdi Main is weakest on every measure:** lowest revenue (8.0% share, only 52% of Abuja Wuse's), fewest orders (98), fewest units (563) and lowest average order value (NGN 178,945). It sits 36.4% below the network average.
- **Other below-average branches:** Lagos Lekki (-14.7%), Otukpo Branch (-9.8%) and Kano Sabon Gari (-1.0%, effectively level with the average).
- **Port Harcourt is a low-traffic, high-value branch:** only 103 orders (second fewest) but the highest AOV of all branches (NGN 288,666).

Question 2. Which product categories and individual products sell best, by both revenue and quantity?

Answer. Electronics earns the most revenue (NGN 79.3M, 35.9%) from relatively few units, while Home & Living sells the most units (1,863). By product, Foam Mattress (6x6) is #1 by revenue (NGN 28.4M) and Cooking Gas Cylinder 12.5kg is #1 by quantity (527 units).

Category	Gross sales (NGN)	Share	Units sold	Orders	Revenue per unit (NGN)
Electronics	79,250,800	35.9%	1,403	233	56,487
Home & Living	73,926,100	33.5%	1,863	268	39,681
Groceries	44,071,100	20.0%	1,612	219	27,339
Fashion	23,282,000	10.6%	1,388	236	16,774

#	Top 5 by revenue	Gross sales (NGN)	Top 5 by quantity	Units sold
1	Foam Mattress (6x6)	28,410,800	Cooking Gas Cylinder 12.5kg	527
2	Cooking Gas Cylinder 12.5kg	24,226,700	Foam Mattress (6x6)	387
3	LED TV 43-inch	19,292,900	Bedsheet Set	380
4	Rice 50kg Bag	18,540,700	Spaghetti Carton	351
5	Smartphone - Tecno Spark	17,523,400	Standing Fan	333

- **Two categories carry the business:** Electronics and Home & Living together earn 69.5% of revenue. Groceries adds 20.0% (NGN 44.1M) and Fashion is the smallest at 10.6% (NGN 23.3M) despite selling 1,388 units.
- **Price per unit explains the difference:** Electronics earns about NGN 56,487 per unit sold against NGN 39,681 for Home & Living, NGN 27,339 for Groceries and NGN 16,774 for Fashion.
- **Concentration:** the top two products, both Home & Living, generate 23.9% of all revenue. The weakest products by revenue are Detergent 4kg (NGN 2.0M), Blender (NGN 2.1M) and Vegetable Oil 5L (NGN 2.2M).

Question 3. How does monthly sales performance trend across the year? Is there seasonality?

Answer. Sales swing sharply from month to month but trend upward through the year: June is the low (NGN 12.6M) and December the high (NGN 26.8M, 2.1x June). Q4 is the strongest quarter and the second half is 14.0% above the first. December's peak is consistent with holiday-driven demand, although one year of data cannot confirm a repeating seasonal pattern.

Month	Gross sales (NGN)	Orders	Units sold	Change vs previous month
January	21,267,800	98	623	—
February	13,935,100	67	350	-34.5%
March	15,773,100	82	538	+13.2%
April	19,381,800	79	571	+22.9%
May	20,048,000	74	537	+3.4%
June	12,647,200	62	367	-36.9%
July	20,267,500	75	618	+60.3%
August	23,035,100	92	538	+13.7%
September	14,345,600	76	459	-37.7%
October	16,488,700	76	498	+14.9%
November	16,574,000	81	465	+0.5%
December	26,766,100	94	702	+61.5%
FY2024 total	220,530,000	956	6,266	

- **Peak and trough:** December is the highest month by sales and by units sold (702), with 94 orders, second only to January (98). June is the lowest month by sales and has the fewest orders (62).
- **Not a smooth curve:** there are three sharp drops, in February (-34.5%), June (-36.9%) and September (-37.7%), each followed by a rebound. Six months beat the NGN 18.4M monthly average: Jan, Apr, May, Jul, Aug, Dec.
- **Upward drift:** quarterly gross sales rise steadily, Q1 NGN 51.0M, Q2 NGN 52.1M, Q3 NGN 57.6M, Q4 NGN 59.8M; H2 (NGN 117.5M) is 14.0% above H1 (NGN 103.1M).

Question 4. How much revenue is being lost to discounts, and which customer type receives the most discount?

Answer. About NGN 2,178,270 is given away in discounts, just 0.99% of gross sales (NGN 220,530,000), so discounting is a small leak rather than a major drain. Wholesale customers receive the most: NGN 1,454,230, or 66.8% of all discounts.

Customer type	Gross sales (NGN)	Discount given (NGN)	Discount rate	Share of discounts	Orders discounted	Discount per order (NGN)
Wholesale	98,521,800	1,454,230	1.48%	66.8%	35 of 227 (15.4%)	6,406
Retail	103,712,700	608,290	0.59%	27.9%	64 of 644 (9.9%)	945
Corporate	18,295,500	115,750	0.63%	5.3%	10 of 85 (11.8%)	1,362
Total	220,530,000	2,178,270	0.99%	100.0%	109 of 956 (11.4%)	2,279

- **Wholesale is discounted most, in both size and frequency:** its discount rate (1.48% of its gross sales) is about 2.5x Retail's (0.59%), and 15.4% of Wholesale orders carry a discount against 9.9% of Retail orders. Corporate is at 0.63%.
- **Discounts are occasional:** only three discount levels appear in the data (5%, 10% and 15%), and just 109 of 956 orders (11.4%) carry any discount at all.
- **By branch:** Otukpo Branch (1.75%) and Lagos Ikeja (1.55%) discount most heavily; Port Harcourt least (0.28%). **By category:** Groceries (1.47%) and Fashion (1.39%) are discounted most, Electronics least (0.54%).

Question 5. Retail vs Wholesale vs Corporate: which customer segment is most valuable to the business?

Answer. It depends on the yardstick. Retail is the largest segment by total revenue (NGN 103.7M, 47.0%) and by order count (644), but Wholesale is the most valuable per order: NGN 434,017 on average, 2.7x Retail's NGN 161,045. Wholesale earns 95% of Retail's revenue from only 35% as many orders. Corporate is the smallest segment (8.3%).

Segment	Orders	Share of orders	Gross sales (NGN)	Share of sales	AOV, gross (NGN)	AOV, net (NGN)
Retail	644	67.4%	103,712,700	47.0%	161,045	160,100
Wholesale	227	23.7%	98,521,800	44.7%	434,017	427,610
Corporate	85	8.9%	18,295,500	8.3%	215,241	213,879
Total	956	100.0%	220,530,000	100.0%	230,680	228,401

- **Retail** is the volume base: 67.4% of all orders but the lowest AOV. **Wholesale** is the most efficient segment: even after its larger discounts, its net AOV (NGN 427,610) is 2.7x Retail's (NGN 160,100), so one Wholesale order is worth roughly 3 Retail orders.
- **Buying patterns differ:** Wholesale spend is concentrated in Home & Living (39.4%) and Groceries (26.3%); Retail (42.5%) and Corporate (53.0%) lean towards Electronics.

Question 6. Which payment methods are customers actually using, and does this vary by branch?

Answer. Payment methods are spread almost evenly across the network: of 935 orders with a recorded method, POS/Card accounts for 26.2%, Bank Transfer 25.9%, Mobile Money 24.6% and Cash 23.3%. Individual branches lean differently (Lagos Ikeja most towards POS/Card, 36.2%), but with only about 100 to 140 orders per branch these differences are not statistically significant (chi-square test, $p = 0.56$).

Branch	Bank Transfer	Cash	Mobile Money	POS/Card	(blank)	Total	Most used
Abuja Wuse	33	37	37	28	4	139	Cash / Mobile Money (tie)
Gboko Branch	38	33	28	32	2	133	Bank Transfer
Kano Sabon Gari	35	28	29	25	1	118	Bank Transfer
Lagos Ikeja	23	29	29	46	2	129	POS/Card
Lagos Lekki	32	21	31	28	3	115	Bank Transfer
Makurdi Main	25	19	20	31	3	98	POS/Card
Otukpo Branch	32	28	30	27	4	121	Bank Transfer
Port Harcourt	24	23	26	28	2	103	POS/Card
All branches	242	218	230	245	21	956	POS/Card
% of recorded	25.9%	23.3%	24.6%	26.2%			

- **Branch leanings:** POS/Card leads at Lagos Ikeja (36.2% of its recorded payments), Makurdi Main (32.6%) and, narrowly, Port Harcourt (27.7%). Bank Transfer leads at Gboko (29.0%), Kano Sabon Gari (29.9%), Lagos Lekki (28.6%) and Otukpo (27.4%). Abuja Wuse splits Cash and Mobile Money equally (37 orders each).
- **Reading the variation:** branch percentages are based on recorded payments only, and each branch has a small sample, so the leanings above are tendencies rather than firm branch-specific behaviour. 21 orders (2.2%) have no payment method recorded and appear as "(blank)".

Question 7. Are there data quality problems in the export that could be distorting any of the figures above?

Answer. Yes. The raw export contained 15 exact duplicate rows, 32 spellings of 8 branch names and 12 spellings of 4 category names, 6 returns coded as negative quantities, and blanks in Discount %, Payment Method and Salesperson. All were handled in the cleaning step (full detail in section 2.5), so none of them materially distorts the figures reported in Questions 1 to 6.

2.4 Hypotheses tested and validated

Hypothesis	Result
Wholesale customers receive a proportionally larger discount than Retail customers.	Validated. Wholesale's discount rate (1.48% of gross sales) is about 2.5x Retail's (0.59%).
Sales show holiday seasonality, peaking in December.	Supported, with a caveat. December is the highest month (NGN 26.8M), Q4 the strongest quarter and June the lowest month (NGN 12.6M). With a single year of data, a repeating seasonal pattern cannot be confirmed.
The segment with the most orders is also the most valuable per order.	Rejected. Retail has the most orders (644) but the lowest average order value (NGN 161,045); Wholesale has fewer orders (227) but the highest (NGN 434,017).
The state with the most branches (Benue) also has the strongest branches.	Rejected. Benue has the largest state total (NGN 74.7M) but the lowest revenue per branch (NGN 24.9M) and includes the weakest branch, Makurdi Main.

2.5 Data quality issues detected

Issue	Extent	How it was handled	Effect on the results
Exact duplicate rows	15 rows (971 to 956)	Removed; one copy of each transaction kept.	Left in, they would have overstated gross sales by NGN 2,173,500 (1.0%): NGN 222,703,500 instead of NGN 220,530,000.
Inconsistent casing and spacing: Branch	32 spellings of 8 branches	Standardised to one spelling per branch (spaces trimmed, capitalisation made consistent).	Uncleaned, each PivotTable would list the same branch several times and split its totals.
Inconsistent casing and spacing: Category	12 spellings of 4 categories	Standardised to one spelling per category.	Same problem: category totals would be split across variants.
Miscoded returns (negative quantities)	6 rows; -24 units; NGN -1,393,200 gross	Kept and netted into all totals.	Netting returns against sales is standard practice. A "before returns" view of gross sales would read NGN 221,923,200 (0.6% higher).
Missing Discount %	55 rows (5.8%); 56 in the raw export	Left blank; discount totals use the Discount Amount column.	53 of the 55 rows carry no discount amount; the other 2 carry NGN 15,700 in discounts, which is included. Discount totals are unaffected.
Missing Payment Method	21 rows (2.2%)	Left blank; shown as "(blank)" in the payment PivotTable.	Covers NGN 7,247,600 (3.3%) of gross sales. Payment percentages in Question 6 use recorded payments only.
Missing Salesperson	32 rows (3.3%); 33 in the raw export	Left blank.	No effect: salesperson is not used in any of the seven questions.

Note on returns. The 6 return rows (total -24 units, NGN -1,393,200 in gross sales) are netted into every total in the workbook, including the NGN 220,530,000 gross sales figure used throughout this report. They were not stripped out separately; the point worth flagging is only that they are netted in rather than excluded.

TASK 3 · DATA VISUALIZATION

3.1 Tool and approach

Visualization was built natively in Microsoft Excel rather than Matplotlib, Seaborn or Tableau. PivotTables aggregate the cleaned 956-row dataset, PivotCharts visualize each aggregation, and everything is laid out on a single dashboard sheet so the whole FY2024 story reads in one view: five key-figure tiles, six charts and one interactive slicer.

3.2 Where to find the dashboard in the workbook

The dashboard is deliberately not reproduced as a picture in this report, because it is interactive and is meant to be reviewed inside the workbook. Open **Sales_Analysis_Workbook.xlsx** and go to the **Dashboard** tab (titled "IVOM RETAIL SALES PERFORMANCE DASHBOARD"). The four tabs, in order, are:

Tab	What it contains
Raw Sales Data	The original POS export as received (971 rows, 14 columns), kept in its original uncleaned state.
working sheet	The cleaned dataset (956 rows, 15 columns): duplicates removed, Branch and Category standardised, Discount Amount column added. Source of every PivotTable.
pivotable	The 11 PivotTables (6 feed the charts, 5 feed the key-figure tiles) and the six PivotCharts built on them.
Dashboard	The one-sheet dashboard: five key-figure tiles, the six charts and the Order Date slicer.

3.3 Dashboard walkthrough

Key-figure tiles (top row). Each tile is a GETPIVOTDATA formula linked to its own PivotTable, so all five update together when the slicer is used. Full-year values:

Total Gross Sales	Total Net Sales	Total Discount	Total Unit Price	Total Quantity
NGN 220,530,000	NGN 218,351,730	NGN 2,178,270	NGN 35,378,200	6,266 units

Total Unit Price is the sum of the unit-price column and is shown as a reference figure; it is not a measure of business performance.

The six charts (each sits in its own colour-coded tile, so the eye can separate the sections at a glance):

#	Chart	Type and content	What it shows (full-year view)
1	Gross Sales by Branches & State	Stacked column chart with a two-level axis: each branch sits under its state.	Q1. Abuja Wuse is the tallest column (NGN 33.7M) and Makurdi Main the shortest (NGN 17.5M); grouped under their state, the three Benue branches together add up to the largest state total.
2	Gross Sales and Quantity by Products	Combo chart: columns for gross sales and a line for units sold, by category (two units on two axes).	Q2. Electronics earns the most (NGN 79.3M) from 1,403 units; Home & Living sells the most units (1,863).
3	Monthly performance in the Year	Line chart of gross sales, January to December.	Q3. Trough in June (NGN 12.6M), peak in December (NGN 26.8M), with further dips in February and September.
4	Discount	Pie chart of discount amount by customer type, with values and percentages.	Q4. Wholesale takes the largest slice (66.8%), then Retail (27.9%) and Corporate (5.3%).
5	Customer Segment	Doughnut chart of gross sales by customer type, with values and percentages.	Q5. Retail (47.0%) and Wholesale (44.7%) dominate; Corporate is a distant third (8.3%).
6	Payment Method	Stacked column chart of order counts by payment method for each branch, with data labels.	Q6. The four methods are fairly balanced at most branches; Lagos Ikeja has the largest POS/Card segment (46 orders).

3.4 Using the interactive slicer

An **Order Date** slicer (one copy on the pivotable tab and one on the Dashboard tab, linked to each other) is connected to all 11 PivotTables. Clicking a month re-cuts every tile and chart at once, for example selecting December shows December only. To return to the full-year figures quoted in this report, click the slicer's **Clear Filter** button (the funnel icon at its top right). Slicers exist only in Excel, so they cannot appear in a PDF; this is the main reason the dashboard should be viewed in the workbook itself.

3.5 Design choices

- **One sheet, one story:** key figures on top for quick reference, then the charts arranged in a grid beneath, so a reviewer can read the whole year without switching sheets.
- **Chart type matched to the question:** columns for comparing branches, a line for a trend over time, pie and doughnut for share of total, and a combo chart where two different units (naira and units sold) need to appear together.
- **Colour-coded tiles:** each chart sits on its own background colour, so the sections are visually separated and easy to find.
- **Interactive by design:** because every tile and chart reads from PivotTables tied to one slicer, the dashboard can be re-cut by month without rebuilding anything.